Shipment Signal Control Tower: Complete Master Project Guide

Hackathon: Cognizant MuleSoft Hackathon 2026–2027

Track: Use Case 08: Logistics – Real-Time Shipment Tracking

Application Name: `shipment-tracking-control-tower`

Application Directory: `Cognizant-Hackathon/Shipment-Signal-Control-Tower/`

Guides Directory: `Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/`

Clickable Table of Contents

1. [00 – Start Here: Onboarding & Roadmap]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/00-START-HERE.md)

2. [01 – Project Overview & Business Context]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/01-Project-Overview.md)

3. [02 – Software Installation & Setup Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/02-Software-Installation-and-Setup.md)

4. [03 – Anypoint Studio Project Import & Execution]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-

Hackathon/Shipment-Tracking-Complete-Guide/03-Anypoint-Studio-Project-Import.md)

5. [04 – Architecture & API-Led Connectivity]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/04-Architecture-and-API-Led-Connectivity.md)

6. [05 – RAML 1.0 & APIKit Complete Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/05-RAML-and-APIKit-Complete-Guide.md)

7. [06 – All Mule Flows Explained]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/06-All-Mule-Flows-Explained.md)

8. [07 – DataWeave 2.0 & Carrier Mappings]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/07-DataWeave-and-Carrier-Mappings.md)

9. [08 – MySQL Database Setup & Management]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/08-MySQL-Database-Setup.md)

10. [09 – Authentication & Security Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/09-Authentication-and-Security.md)

11. [10 – Postman Complete Testing Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/10-Postman-Complete-Testing-Guide.md)

12. [11 – MUnit Complete Testing Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/11-MUnit-Complete-Testing-Guide.md)

13. [12 – Error Handling, Resilience & Retry]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/12-Error-Handling-and-Retry.md)

14. [13 – Queue & Notification Architecture]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/13-Queue-and-Notification-Guide.md)

15. [14 – Anypoint Platform & CloudHub Deployment]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/14-Anypoint-Platform-and-CloudHub-Deployment.md)

16. [15 – Logging, Observability & Troubleshooting]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/15-Logging-and-Troubleshooting.md)

17. [16 – Hackathon Live Demonstration Guide]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/16-Hackathon-Demo-Guide.md)

18. [17 – Judge Questions & Answers Architectural Defense]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/17-Judge-Questions-and-Answers.md)

19. [18 – Six-Member Team Responsibilities & RACI Matrix]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/18-Six-Member-Team-Responsibilities.md)

20. [19 – Requirements Traceability & Verification Checklist]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/19-Requirements-and-Verification-Checklist.md)

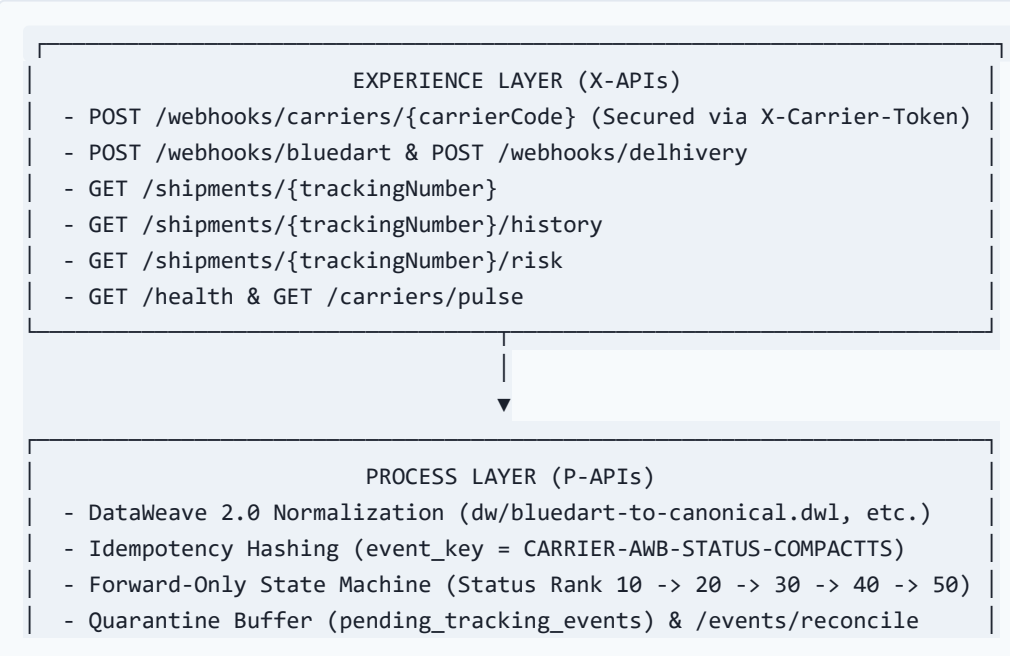
21. [20 – Project Limitations & Future Scope Roadmap]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/20-Project-Limitations-and-Future-Scope.md)

22. [Final Status Report]
(file:///c:/Users/DELL/OneDrive/Desktop/Hackathon11/Cognizant-Hackathon/Shipment-Tracking-Complete-Guide/FINAL-STATUS-REPORT.md)

1. Executive Master Summary

The **Shipment Signal Control Tower** is a production-grade, event-driven logistics integration application engineered natively on MuleSoft Mule 4. It integrates with multi-carrier webhook push interfaces (supporting **BlueDart Express** and **Delhivery Logistics**), normalizes diverse schemas into a unified **Canonical Lifecycle Event Model**, enforces strict **idempotency**, guarantees **forward-only shipment status progression**, quarantines unmanifested tracking events with automated batch reconciliation, calculates an **Explainable Delay Risk Score** against promised customer delivery SLAs, and guarantees zero message loss using the **Transactional Notification Outbox Pattern** and persistent message queues.

2. System Architecture Blueprint



```
│  - Explainable Delay Risk Engine (ON_TRACK, AT_RISK, DELAYED, CRITICAL)│
│  - Milestone Notification Filter (PICKED_UP, OFD, DELIVERED, EXCEPTION)│
└──────────────────────────────────────┬──────────────────────────────────┘
                                       │
                                       ▼
┌──────────────────────────────────────────────────────────────────────────┐
│                        SYSTEM LAYER (S-APIs)                             │
│  - MySQL 8.0 Relational Persistence (Database: shipment_control_tower)  │
│    ├── shipments (Master state & SLA risk)                               │
│    ├── shipment_tracking_events (Immutable chronological audit trail)   │
│    ├── pending_tracking_events (Quarantine buffer for unmanifested)     │
│    ├── notification_outbox (Transactional outbox for alerts)            │
│    └── integration_alerts (Operational telemetry & SLA breaches)        │
│  - Asynchronous Message Broker (Persistent VM Queues / Anypoint MQ)     │
│    ├── shipment-milestone-queue (FIFO Persistent Queue)                 │
│    └── shipment-milestone-dlq (Dead-Letter Queue)                       │
└──────────────────────────────────────────────────────────────────────────┘
```

3. Implemented Canonical Status Lifecycle

Every carrier event is normalized to one of the 9 official Use Case 08 lifecycle statuses:

```
[LABEL_CREATED] (Rank 10)
       │
       ▼
  [PICKED_UP]   (Rank 20) ──► (Triggers Customer Milestone Alert)
       │
       ▼
  [IN_TRANSIT]  (Rank 30)
       │
       ▼
[ARRIVED_AT_HUB](Rank 35)
       │
       ▼
[OUT_FOR_DELIVERY](Rank 40) ──► (Triggers Customer Milestone Alert)
       │
       ├──────────────────────────────────┐
       ▼                                  ▼
[DELIVERY_ATTEMPTED] (Rank 45)      [EXCEPTION] (Rank 42)
       │                                  │
       ▼                                  ▼
```

```
 [DELIVERED] (Rank 50 - Terminal)  [RETURNED] (Rank 60 - Terminal)
 (Triggers Milestone Alert)        (Triggers Milestone Alert)
```

---

4. End-to-End Execution Sequence

```
HTTP Ingress Listener (/api/v1/webhooks/carriers/{carrierCode})
    │
    ▼
X-Correlation-ID Extraction or Auto-Generation
    │
    ▼
Carrier Authentication Check (X-Carrier-Token)
    │
    ▼
DataWeave 2.0 Normalizer to Canonical Event Model
    │
    ▼
Idempotency Query on Unique event_key in Database
    │
    ├── IF Duplicate Found:
    │   └── Suppress DB update & notifications -> Return HTTP 200 DUPLICATE_ACKNOWLEDGED
    │
    └── ELSE (Fresh Event):
        └── Master Shipment Lookup by tracking_number
            │
            ├── IF Unmanifested / Not Found:
            │   └── Insert into pending_tracking_events -> Return QUARANTINED_PENDING_REGI
            │
            └── IF Found:
                └── Evaluate Rank & Chronology
                    │
                    ├── IF Out-of-Order / Stale:
                    │   └── Insert to audit history as OUT_OF_ORDER_RECORDED
                    │   └── Do NOT regress master status -> Return OUT_OF_ORDER_RECORDED
                    │
                    └── ELSE (Normal Forward Progression):
                        └── Compute Risk Score & SLA Breach (risk-assessment.xml)
                        └── Update shipments master record
                        └── Insert into shipment_tracking_events as PROCESSED
                        └── Filter Milestone Notification
                            │
                            └── IF Milestone Eligible:
                                └── Insert into notification_outbox table (Status: QUEUED)
```

```
                                    └── Publish to VM Queue (shipment-milestone-queue)
                                    └── Asynchronous Consumer Dispatches & Marks Outbox SENT
                            └── Return HTTP 200 EVENT_PROCESSED_SUCCESSFULLY
```

---

5. Converting This Guide to PDF (`COMPLETE-PROJECT-GUIDE.pdf`)

To generate `COMPLETE-PROJECT-GUIDE.pdf` from this Markdown file, you can use any of the following standard approaches:

Method 1: Using Node.js `md-to-pdf` or `markdown-pdf` (Recommended)

```
# In terminal:
npx md-to-pdf COMPLETE-PROJECT-GUIDE.md
```

Method 2: Using VS Code / IDE Markdown PDF Extension

1. Open `COMPLETE-PROJECT-GUIDE.md` in any markdown editor (VS Code, Obsidian, Typora).

2. Right-click and select **Markdown PDF: Export (pdf)**.

Method 3: Using Chrome / Edge Browser Print to PDF

1. Open this file in any markdown viewer or browser.

2. Press `Ctrl + P` -> Select **Destination: Save as PDF** -> Click **Save**.